# Motion in a straight line CHEAT SHEET



## POSITION, DISPLACEMENT, VELOCITY

frame of reference: where motion is measured from

position x: metres from the origin; + one way, − the other

path length: every step added (always ≥ 0)

displacement $\Delta x = x_2 - x_1$: 0 → 40 → 10 m: +10 m, path 70 m

average velocity $\frac{\Delta x}{\Delta t}$: 10 m in 50 s = 0.2 m/s

average speed $\frac{\text{path length}}{\Delta t}$: 70 m in 50 s = 1.4 m/s

round trip: velocity 0, speed not; km/h ÷ 3.6 = m/s

## EDGE TESTS

a = 0: v = u, x = ut (a straight x–t line)

t = 0: v = u, x = 0

from rest: x grows as $t^2$ (1 : 4 : 9 after 1, 2, 3 s)

double u → four times the braking distance (10 → 40 m)

v = 0 does not mean a = 0 (the top of a throw)

## THE GRAPH CHAIN: SLOPES AND AREAS

slopes go DOWN the chain: x–t → v–t → a–t

v = slope of x–t = $\frac{dx}{dt}$ a = slope of v–t = $\frac{dv}{dt}$

$x = 1.5t^2$: windows from 2 s → 7.5, 6.15, 6.015 → 6 m/s

areas go UP the chain: a–t → v–t → x–t

area under v–t = displacement; under a–t = Δv

strips under 0 → 10 m/s in 4 s: 15, 19.5, 19.95 → 20 m

area below the axis counts backwards: +20 − 10 = 10 m

the lift: 2 + 20 + 2 = 24 m (8 floors)

## THE THREE EQUATIONS (CONSTANT A)

constant a only: u start, v later, t time, x displacement

$v = u + at$ (slope of v–t) no x

$x = ut + \frac{1}{2}at^2$ (rectangle + triangle) no v

$v^2 = u^2 + 2ax$ (trapezium, t removed) no t

$x = \frac{u + v}{2} t$ (average of u and v, times t)

car: u = 10, a = 2, t = 5 → v = 20 m/s, x = 75 m

braking from u to rest: $x = \frac{u^2}{2a}$

## THE SAME, BY CALCULUS

power rule: $\frac{d}{dt} t^n = n\, t^{n-1}$; a number → 0

a = dv/dt = const → $v = u + at$

$v = dx/dt = u + at \rightarrow x = ut + \frac{1}{2}at^2$

$a = v \frac{dv}{dx} \rightarrow v\, dv = a\, dx \rightarrow v^2 = u^2 + 2ax$

changing a (a = 3 − 0.5t, 6 s): calculus 36 m; misused 27 m

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## FREE FALL

free fall: $a = -g$ if up is +; $g = 9.8\ m/s^2$

from rest: $v = gt$ (9.8, 19.6, 29.4); $h = \frac{1}{2}gt^2$

$t = \sqrt{(2h/g)}$: 0.8 m → 0.404 s, 3.96 m/s (no mass in it)

each second from rest: 4.9, 14.7, 24.5 m (1 : 3 : 5)

thrown up at u: top at u/g, height $\frac{u^2}{2g}$, back at 2u/g

u = 14.7 m/s: 1.5 s, 11.0 m, 3 s; Moon (1.62): ≈ 67 m

## REACTION AND STOPPING

stopping = reaction + braking = $u\,t_r + \frac{u^2}{2a}$

20 m/s, 0.7 s, 7 $m/s^2$: 14 + 28.6 = 42.6 m

ruler drop: 20 cm → $t = \sqrt{(2 \times 0.2/9.8)} \approx 0.20$ s

the van, 25 m to go: 1.5 s to react, brakes 6.5 $m/s^2$

50 km/h: 20.8 + 14.8 = 35.7 m: reaches it at ≈ 42 km/h

30 km/h: 12.5 + 5.3 = 17.8 m: stops, 7.2 m to spare

## CHECK BEFORE YOU TRUST IT

before you trust an answer:

1 which way is positive? (signs on u, v, a, g)

2 is the acceleration constant here? (else: phases, or calculus)

3 do the units match? (m, s, m/s, $m/s^2$; km/h ÷ 3.6)

## RELATIVE VELOCITY

$v_{BA} = v_B - v_A$: B's velocity seen from A

same way: 25 − 20 = 5 m/s; opposite: −25 − 20 = −45 m/s

overtake: gain 40 m at 5 m/s → 8 s (car goes 200 m)